

ARVAIKHEER, Mongolia

WMO: 442880

Lat: 46.2592N

Lon: 102.7889E

Elev: 5948

StdP: 11.80

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-10.5	-26.1	1.6	-7.6	-22.6	2.0	-4.5	30.2	3.9	26.2	5.6	5.2	320	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	81.8	58.3	78.5	57.0	75.4	56.1	62.5	76.1	60.5	72.9	58.7	70.5	10.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	89.9	67.3	55.9	83.1	65.4	54.0	77.5	64.3	31.4	75.8	29.7	73.0	28.3	70.3	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	23.6	21.6	DB	-18.0	87.1	4.9	3.3	-21.6	89.4	-24.5	91.4	-27.2	93.2	-30.8	95.6	(4)
(5)			WB	-18.3	64.9	4.9	3.8	-21.8	67.6	-24.7	69.9	-27.4	72.0	-31.0	74.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	6.3	11.7	24.1	38.8	49.7	59.8	64.2	60.8	50.9	36.8	20.7	10.4	(6)
(7)		DBStd	22.10	9.37	10.18	10.96	9.89	9.48	6.95	6.03	6.06	8.42	9.34	10.53	8.95	(7)
(8)		HDD50	6328	1354	1072	802	351	124	9	0	4	88	415	880	1229	(8)
(9)		HDD65	10596	1819	1492	1267	786	479	184	89	157	424	875	1330	1694	(9)
(10)		CDD50	1329	0	0	0	14	115	302	439	338	116	5	0	0	(10)
(11)		CDD65	123	0	0	0	0	5	26	63	27	2	0	0	0	(11)
(12)		CDH74	970	0	0	0	0	43	252	456	202	17	0	0	0	(12)
(13)		CDH80	178	0	0	0	0	2	44	103	29	0	0	0	0	(13)
(14)	Wind	WSAvg	8.3	6.1	7.1	9.3	11.2	11.2	9.7	7.7	7.3	8.4	8.2	7.1	6.1	(14)
(15)	Precipitation	PrecAvg	9.2	0.0	0.0	0.2	0.3	0.6	1.4	3.2	2.4	0.7	0.2	0.1	0.1	(15)
(16)		PrecMax	15.7	0.4	0.3	0.7	1.2	2.2	5.1	7.1	6.0	2.8	1.1	0.5	0.4	(16)
(17)		PrecMin	3.9	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.8	0.1	0.1	0.2	0.3	0.5	0.9	1.7	1.4	0.6	0.3	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	39.4	56.2	68.3	79.3	84.6	88.0	83.6	76.7	63.7	50.7	35.9	(19)
(20)			MCWB	26.3	29.7	39.1	46.6	51.6	57.7	63.3	60.1	54.5	44.2	37.0	26.6	(20)
(21)		2%	DB	28.7	34.4	49.9	63.2	74.1	80.2	82.7	79.2	71.9	58.7	44.6	31.0	(21)
(22)			MCWB	22.2	25.9	35.2	43.5	50.3	56.5	60.2	57.9	51.6	42.5	33.3	24.0	(22)
(23)		5%	DB	24.2	30.5	45.1	59.1	69.7	76.5	79.0	75.6	68.2	54.9	39.9	27.0	(23)
(24)			MCWB	18.7	23.3	32.5	41.6	48.3	54.9	58.8	57.0	50.2	40.3	30.3	21.0	(24)
(25)		10%	DB	20.2	26.8	40.4	54.9	66.0	73.0	75.5	72.2	64.6	51.0	35.8	23.3	(25)
(26)			MCWB	15.7	20.7	29.8	39.5	46.6	53.8	57.9	55.6	48.5	38.2	27.7	18.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.8	30.2	40.5	48.8	56.7	61.6	66.7	64.0	57.6	47.4	37.5	27.2	(27)
(28)			MCD B	32.9	38.9	52.9	64.5	71.1	77.8	81.5	77.5	71.0	59.8	49.5	34.8	(28)
(29)		2%	WB	22.6	26.2	36.0	44.8	52.9	59.2	63.7	61.4	54.2	43.5	33.7	24.1	(29)
(30)			MCD B	28.9	33.5	48.6	59.7	68.3	74.4	77.6	74.0	66.6	56.8	43.9	30.7	(30)
(31)		5%	WB	18.8	23.5	32.9	42.3	50.2	57.4	61.7	59.4	52.0	41.1	30.7	21.2	(31)
(32)			MCD B	23.8	30.1	44.2	57.1	66.6	70.6	74.3	70.9	65.1	53.1	39.3	26.6	(32)
(33)		10%	WB	15.6	20.8	30.1	40.0	47.9	55.7	59.8	57.6	49.9	38.7	27.8	18.3	(33)
(34)			MCD B	20.1	26.7	40.1	53.7	63.8	68.4	71.3	68.4	62.7	50.3	35.4	23.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.6	17.5	19.1	19.6	19.5	18.6	16.8	17.3	18.4	18.1	16.8	15.2	(35)
(36)			MCDBR	20.4	21.5	23.4	24.0	23.3	22.9	21.6	21.8	21.8	20.1	19.4	(36)	
(37)			MCWBR	15.7	15.9	14.6	13.5	11.6	10.2	9.1	9.8	10.8	12.7	13.5	14.9	(37)
(38)		5% WB	MCDBR	20.0	20.8	22.8	22.5	22.3	20.2	18.9	18.4	20.1	21.0	19.5	19.3	(38)
(39)			MCWBR	15.8	15.7	14.8	13.6	12.5	10.3	9.9	9.6	11.3	13.0	13.5	15.0	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.227	0.253	0.297	0.338	0.352	0.365	0.358	0.334	0.300	0.274	0.244	0.228	(40)
(41)		taud		2.413	2.358	2.285	2.232	2.249	2.307	2.378	2.417	2.458	2.458	2.438	2.403	(41)
(42)		Ebn at Noon		288	298	296	291	290	285	286	289	292	286	277	274	(42)
(43)		Edh at Noon		21	28	35	41	42	40	37	34	30	25	21	19	(43)
(44)	All-Sky Solar Radiation	RadAvg		640	991	1455	1869	2136	2115	2004	1818	1568	1099	698	523	(44)
(45)		RadStd		42	48	55	69	109	123	102	108	67	46	45	30	(45)

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-5.69	N/A	-2.47	-3.26	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	-4.54	N/A	-1.40	-1.57	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page